ECE 340: Semiconductor Devices

Lecture 21: P-N Junction

Prof. Hyunseok Kim

Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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Office Hours: Thursdays 12p-1p, 2256 HMNTL

ECE 340 Midterm I

- Thursday October 15th
  - 7-8 pm, Arrive at Least 15 Minutes Early, Sit Every Other Seat
  - ECEB 1002, 1013, and 1015 – watch for messages on room assignments
- Conflicts
  - Conflict Defined Per UIUC Student Code
  - Must Contact TA Zetai Liu by Friday 10/9 at 5 pm
- DRES
  - Students Must Take Exams at TAC
  - Point of Contact Regardless of Section is John Dallesasse
  - Follow Instructions on Scheduling Provided Earlier
  - Must Provide Date/Time of Exam to TA Zetai Liu by Friday 10/9 at 5 pm
- Illness
  - See Policy in Syllabus, Excused Absence Letter from Dean's Office Required

Exam Location

Section A (Dallesasse): 1002 ECEB

Section B (Gilbert): 1002 ECEB

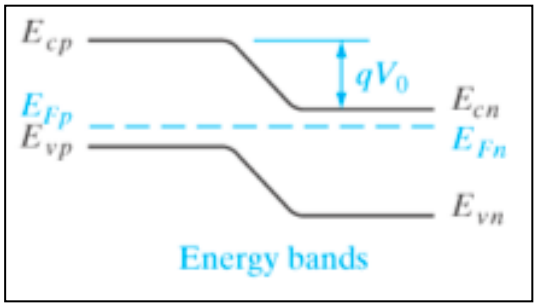
Section C (Kim): 1013 ECEB

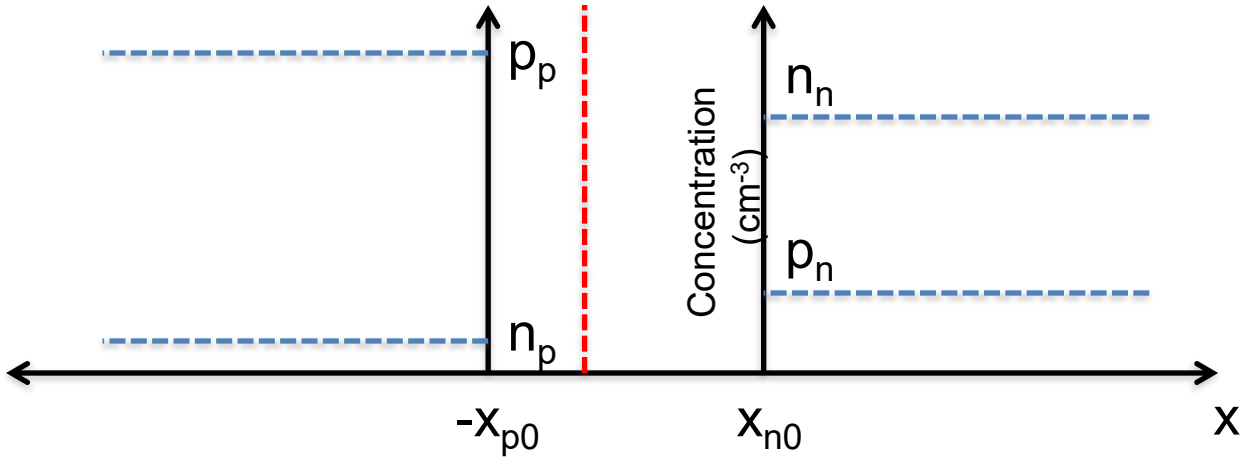
Section D (He): 1015 ECEB

Thursday October 15th

7-8 pm (arrive early)

(Recap) Carrier Concentrations vs. Built-in potential



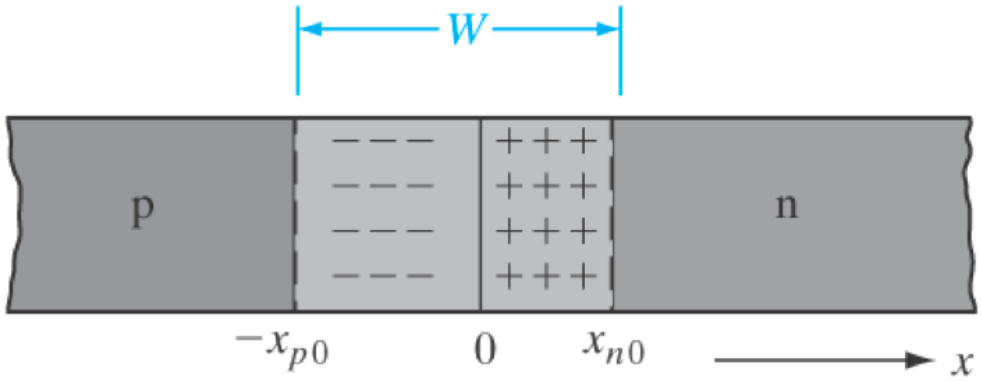


$$\frac{p_p}{p_n} = \frac{n_n}{n_p} = e^{qV_o/kT}$$

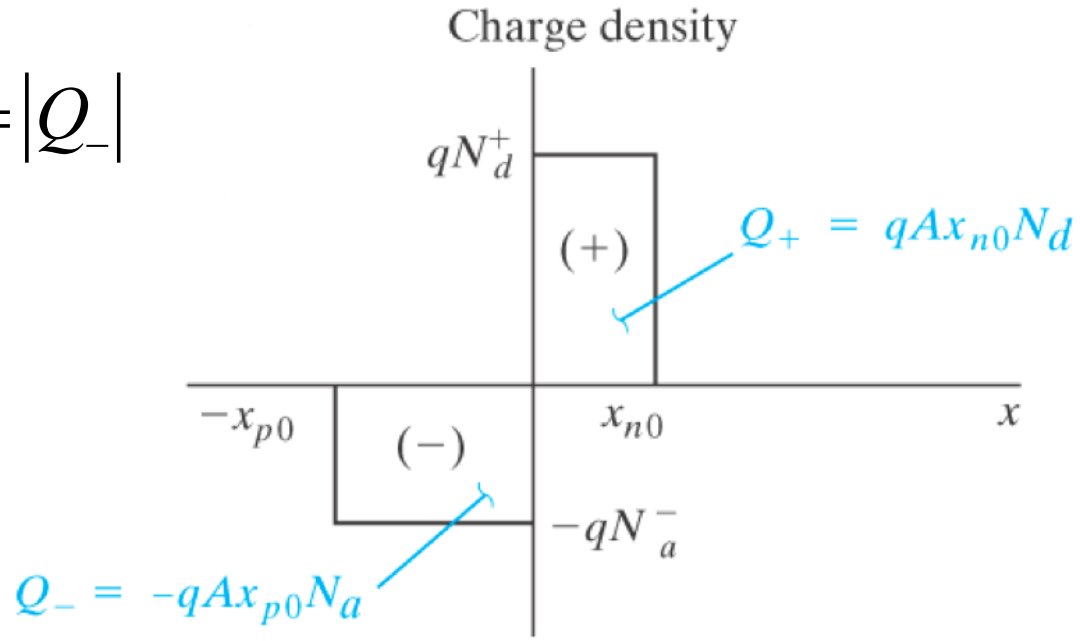
$$p_p = p_n e^{qV_o/kT} \quad \text{or} \quad p_n = p_p e^{-qV_o/kT}$$

$$n_p = n_n e^{-qV_o/kT} \quad \text{or} \quad n_n = n_p e^{qV_o/kT}$$

Space Charge



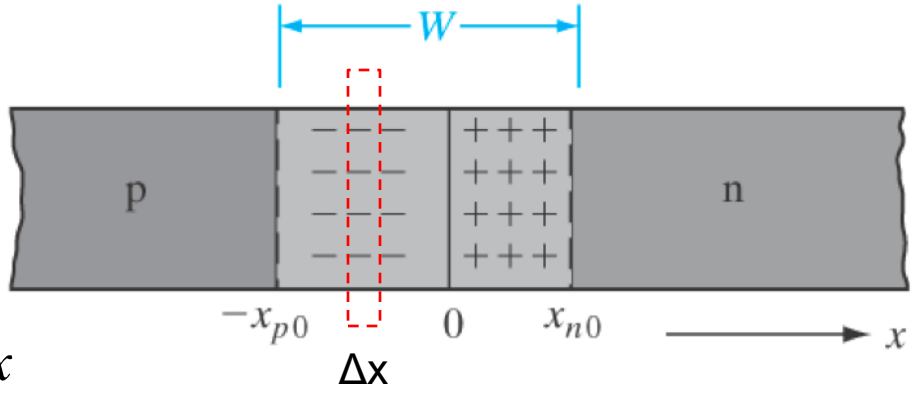
$$|Q_+| = |Q_-|$$



(Recap) Poisson's Equation

Apply Gauss's Law: $\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q}{\varepsilon}$

$$\varepsilon \oint_s \mathcal{E} \bullet dS = \sum_i q_i$$



For a small volume $\Delta x \bullet A$:

$$\varepsilon \cdot \Delta\mathcal{E} \cdot A = q\left[p + N_d^+ - n - N_a^- \right] \cdot A \cdot \Delta x$$

$$\frac{\Delta\mathcal{E}}{\Delta x} = \frac{q}{\varepsilon}\left[p + N_d^+ - n - N_a^- \right]$$

$$\lim_{x \to 0} \frac{\Delta\mathcal{E}}{\Delta x} = \frac{d\mathcal{E}(x)}{dx} = \frac{q}{\varepsilon}\left[p + N_d^+ - n - N_a^- \right]$$

Poisson's Equation:

$$\frac{d\mathcal{E}(x)}{dx} = \frac{q}{\varepsilon}\left[p + N_d^+ - n - N_a^- \right]$$

Calculation of Electric Field: n-Side

n-side of pn junction $(0 < x < x_{no})$:

$$p \approx 0,\ n \approx 0,\ N_a^- \approx 0,\ \frac{d\mathcal{E}(x)}{dx} = \frac{q}{\varepsilon}\left[N_d^+\right]$$

$$\int_{\mathcal{E}_o}^{0} d\mathcal{E}(x) = \frac{q}{\varepsilon}\left[N_d^+\right] \int_{0}^{x_{no}} dx$$

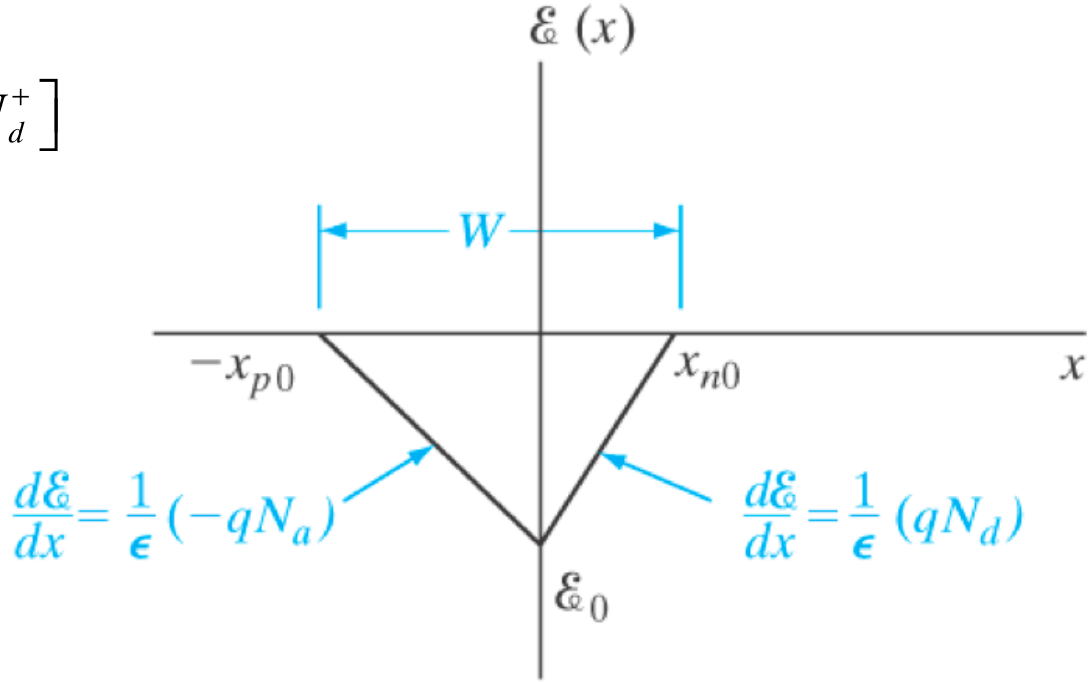
$$\left[0 - \mathcal{E}_o\right] = \frac{q}{\varepsilon}\left[N_d^+\right]\left[x_{no} - 0\right]$$

$$\mathcal{E}_o = -\frac{q}{\varepsilon}\left[N_d^+\right] x_{no}$$

To determine the value $\mathcal{E}(x)$:

$$\int_{\mathcal{E}(x)}^{0} d\mathcal{E}(x) = \frac{q}{\varepsilon}\left[N_d^+\right] \int_{x}^{x_{no}} dx$$

$$\mathcal{E}(x) = -\frac{q}{\varepsilon}\left[N_d^+\right] (x_{no} - x)$$



Maximum Value of Electric Field:

$$\mathrm{E_o} = -\frac{q}{\varepsilon} N_d x_{n0} = -\frac{q}{\varepsilon} N_a x_{p0}$$

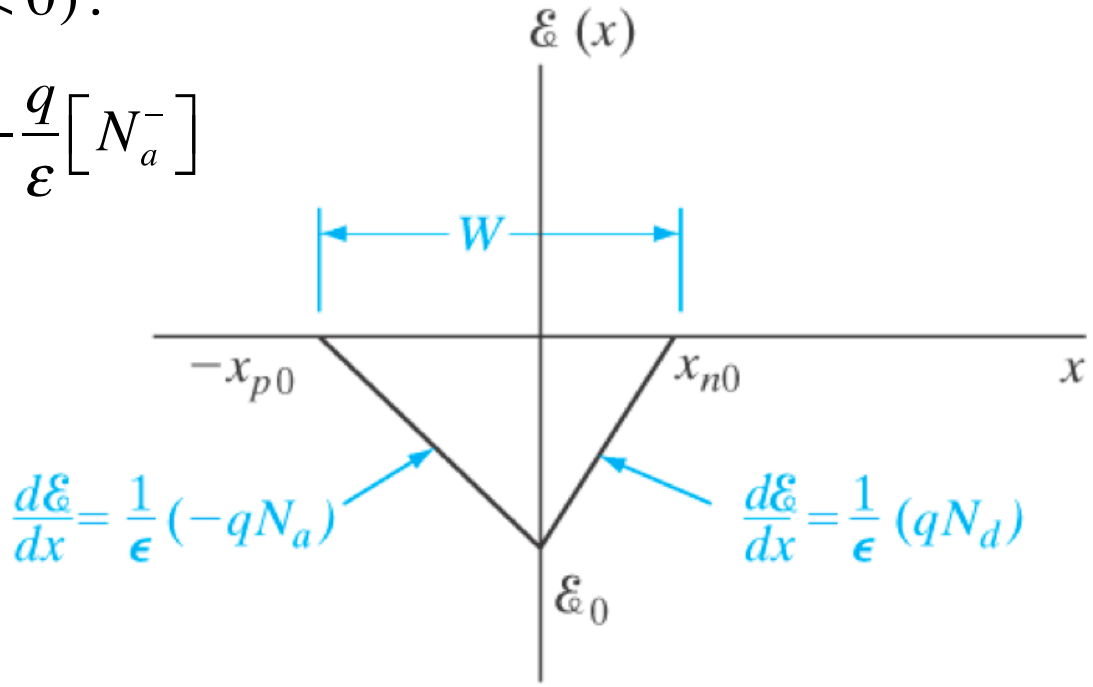
Calculation of Electric Field: p-Side

p-side of pn junction $(-x_{po} < x < 0)$:

$$p \approx 0,\ n \approx 0,\ N_d^+ \approx 0,\ \frac{d\mathcal{E}(x)}{dx} = -\frac{q}{\varepsilon}\left[N_a^-\right]$$

$$\int_0^{\mathcal{E}_o} d\mathcal{E}(x) = -\frac{q}{\varepsilon}\left[N_a^-\right] \int_{-x_{po}}^{0} dx$$

$$\mathcal{E}_o = -\frac{q}{\varepsilon}\left[N_a^-\right] x_{po}$$



To Determine $\mathcal{E}(x)$:

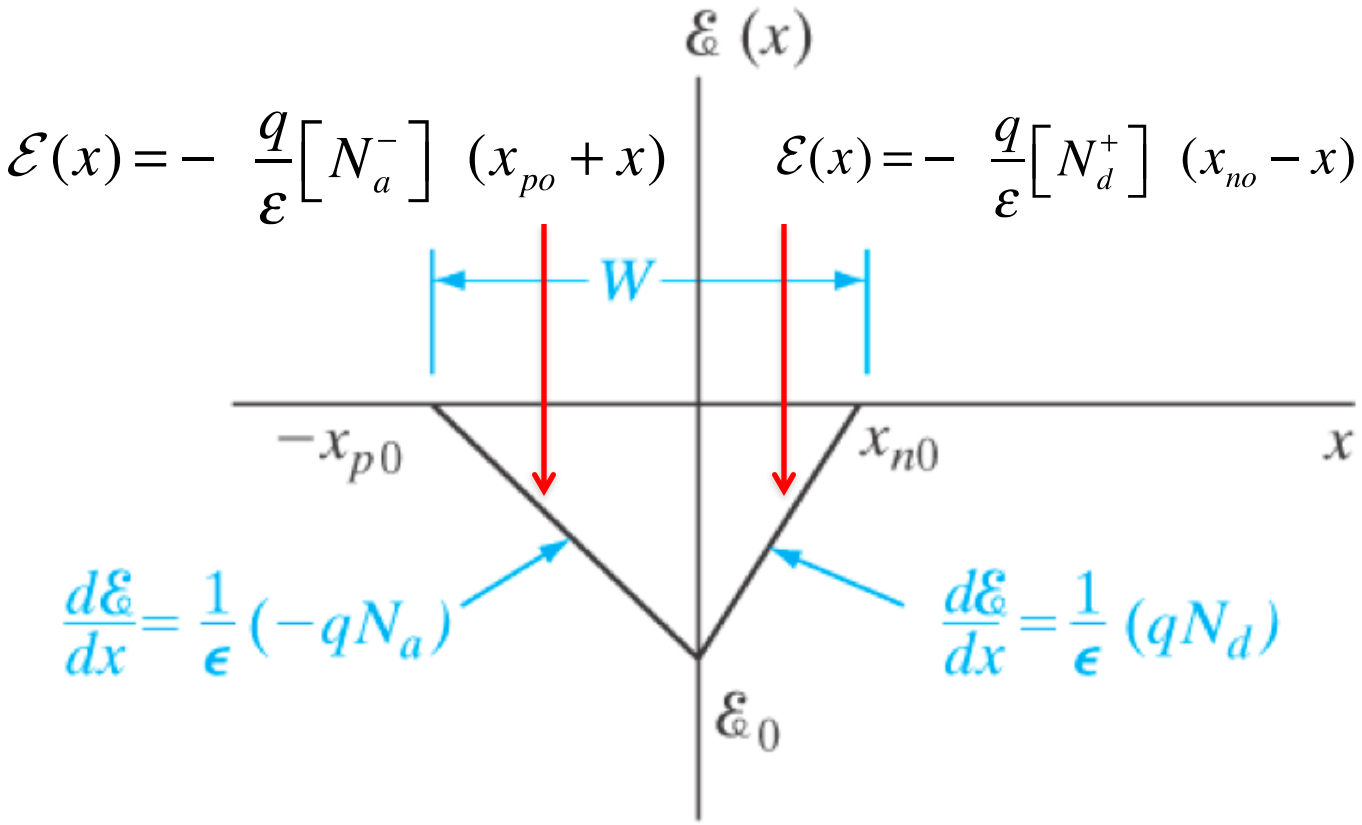
$$\int_0^{\mathcal{E}(x)} d\mathcal{E}(x) = -\frac{q}{\varepsilon}\left[N_a^-\right] \int_{-x_{po}}^{x} dx$$

$$\mathcal{E}(x) = -\frac{q}{\varepsilon}\left[N_a^-\right] (x_{po} + x)$$

Maximum Value of Electric Field:

$$\mathrm{E_o} = -\frac{q}{\varepsilon} N_d x_{n0} = -\frac{q}{\varepsilon} N_a x_{p0}$$

Junction Electric Field: Summary



$$\mathcal{E}_o = -\frac{q}{\varepsilon} N_d x_{n0} = -\frac{q}{\varepsilon} N_a x_{p0}$$

From Integrating Poisson's Equation

Electric Field and Contact Potential

Maximum Value of Electric Field:

$$\mathcal{E}_o = -\frac{q}{\varepsilon} N_d x_{n0} = -\frac{q}{\varepsilon} N_a x_{p0}$$

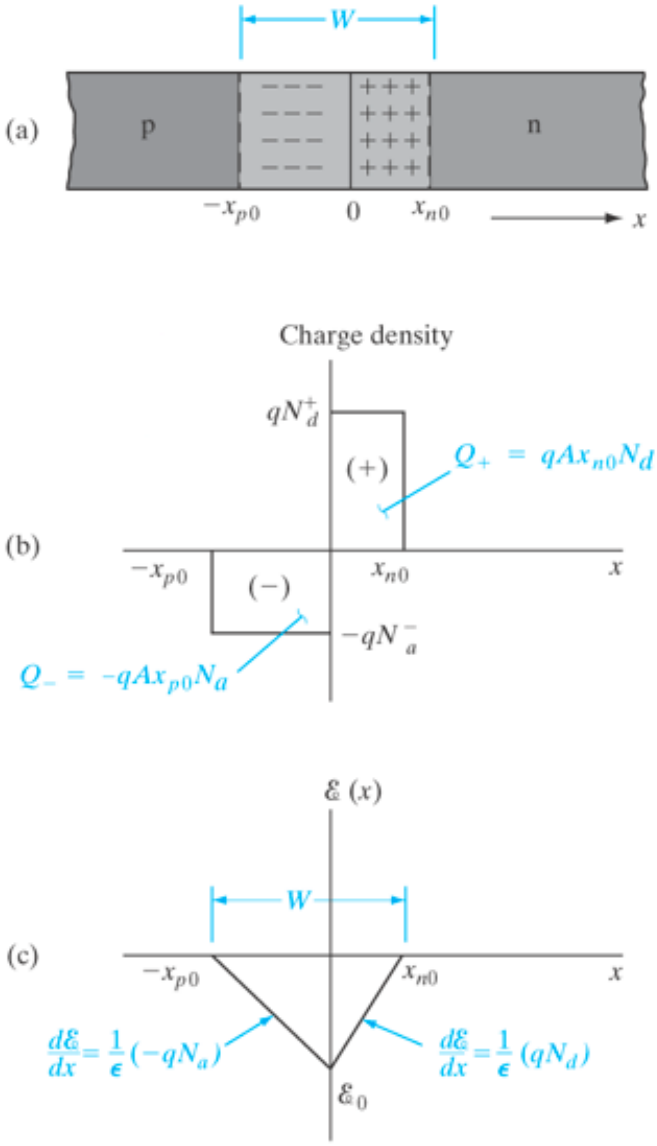
Relationship of Electric Field to Contact Potential:

$$\mathcal{E}(x) = -\frac{d\mathcal{V}(x)}{dx} \text{ or } -V_o = \int_{-x_{po}}^{x_{n0}} \mathcal{E}(x)\,dx$$

Since the electric field is the negative of the potential gradient at any point.

Recall that:

$$V_o = \frac{kT}{q} \ln \frac{p_p}{p_n} = \frac{kT}{q} \ln \frac{n_n}{n_p}$$



Calculation of Depletion Width

From the previous slide:

$$V_o = -\int_{-x_{po}}^{x_{n0}} \mathcal{E}(x)\,dx$$

So, the negative of the contact potential is the area under the $\mathcal{E}$(x) versus x triangle:

$$V_o = -\frac{1}{2}\mathcal{E}_o \cdot W = \frac{1}{2}\frac{q}{\varepsilon}N_d x_{no} W = \frac{1}{2}\frac{q}{\varepsilon}N_a x_{po} W$$

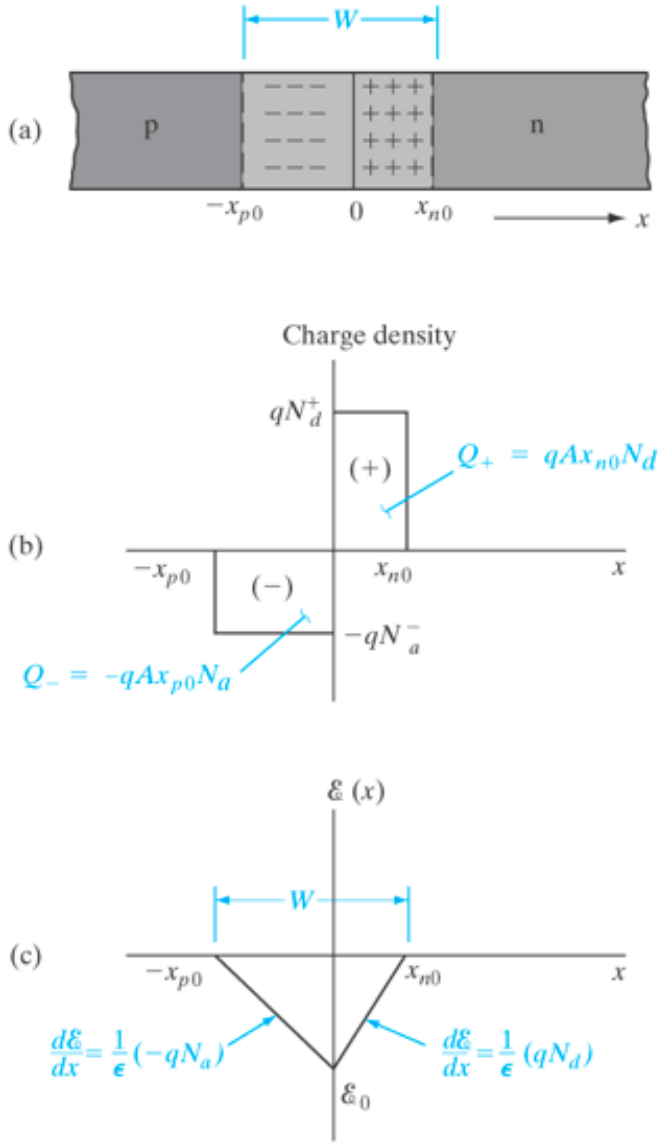
Balance of Charge Requirement: $Q_+ = |Q_-| \Leftrightarrow N_d x_{no} = N_a x_{po}$

$$x_{po} = x_{no}\frac{N_d}{N_a} \text{ and } W = x_{no} + x_{po} = x_{no}\left(1 + \frac{N_d}{N_a}\right),$$

$$\text{so } x_{no} = W\left(\frac{N_a}{N_a + N_d}\right)$$

$$\text{and } V_o = \frac{1}{2}\mathcal{E}_o W = \frac{1}{2}\frac{q}{\varepsilon}\frac{N_a N_d}{N_a + N_d}W^2$$

$$W = \left[\frac{2\varepsilon V_o}{q}\left(\frac{N_a + N_d}{N_a N_d}\right)\right]^{1/2} = \left[\frac{2\varepsilon V_o}{q}\left(\frac{1}{N_a} + \frac{1}{N_d}\right)\right]^{1/2}$$



Other Useful Relationships

$$W = \left[\frac{2\varepsilon V_o}{q}\left(\frac{N_a + N_d}{N_a N_d}\right)\right]^{1/2} = \left[\frac{2\varepsilon V_o}{q}\left(\frac{1}{N_a} + \frac{1}{N_d}\right)\right]^{1/2}$$

and $V_o = \frac{kT}{q}\ln\frac{N_a N_d}{n_i^2}$ so:

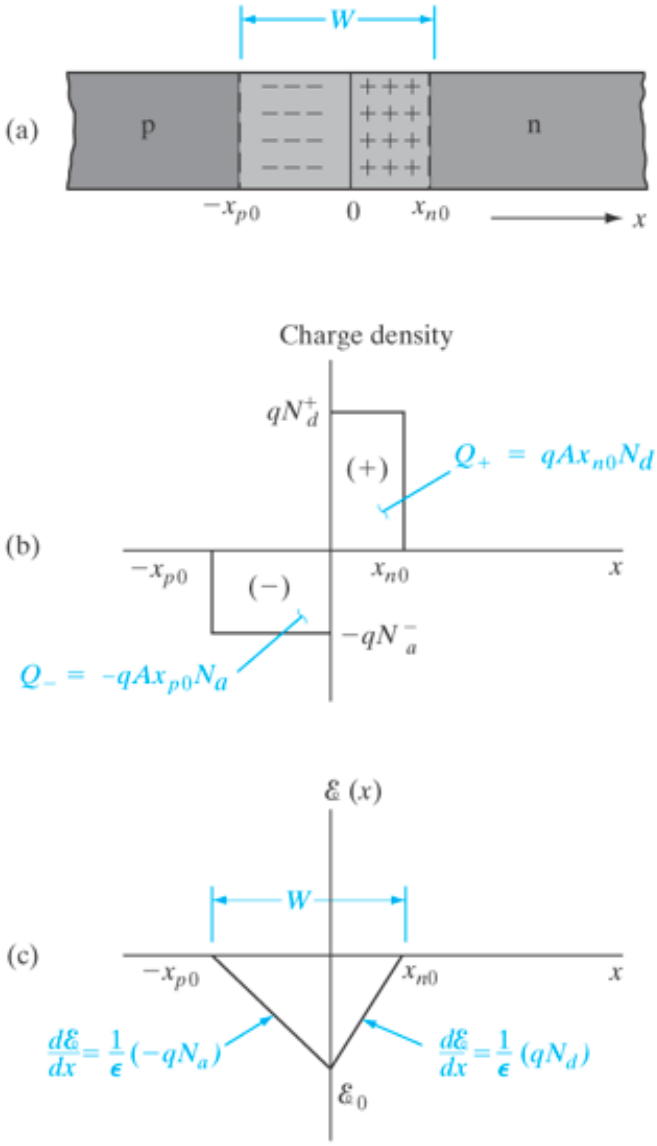
$$W = \left[\frac{2\varepsilon kT}{q^2}\left(\ln\frac{N_a N_d}{n_i^2}\right)\left(\frac{1}{N_a} + \frac{1}{N_d}\right)\right]^{1/2}$$

Also:

$$x_{po} = \frac{WN_d}{N_a + N_d} = \frac{W}{1 + N_a/N_d} = \left\{\frac{2\varepsilon V_o}{q}\left[\frac{N_d}{N_a(N_a + N_d)}\right]\right\}^{1/2}$$

and

$$x_{no} = \frac{WN_a}{N_a + N_d} = \frac{W}{1 + N_d/N_a} = \left\{\frac{2\varepsilon V_o}{q}\left[\frac{N_a}{N_d(N_a + N_d)}\right]\right\}^{1/2}$$

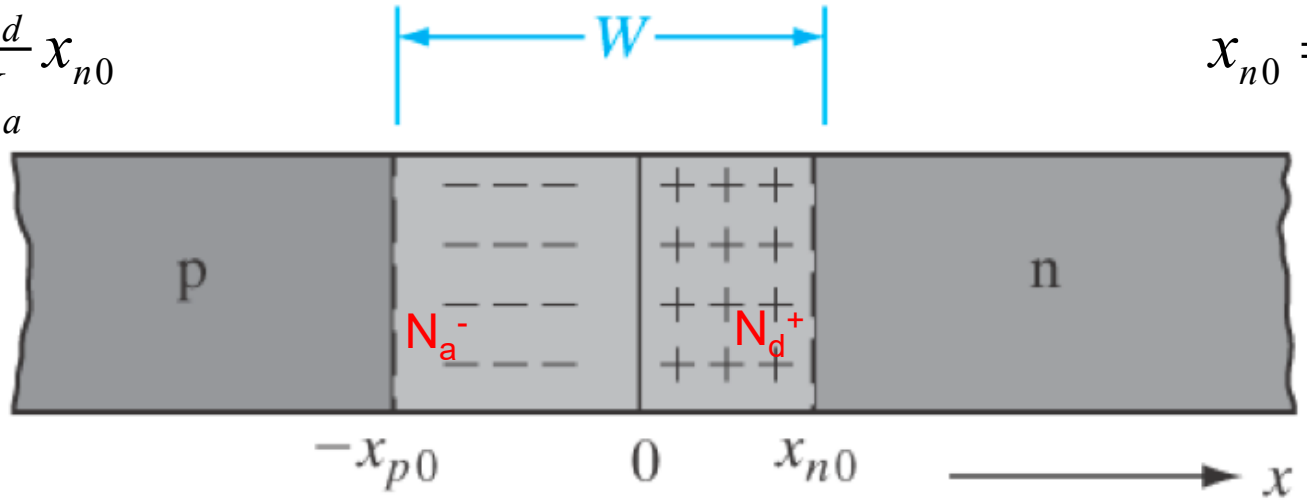


Depletion Width and Contact Potential

$$W = \left[\frac{2\varepsilon V_0}{q} \left(\frac{N_a + N_d}{N_a N_d} \right) \right]^{1/2}$$

$$x_{p0} = \frac{N_d}{N_a} x_{n0}$$

$$x_{n0} = \frac{N_a}{N_d} x_{p0}$$



$$x_{p0} = \left\{ \frac{2\varepsilon V_0}{q} \left[\frac{N_d}{N_a (N_a + N_d)} \right] \right\}^{1/2}$$

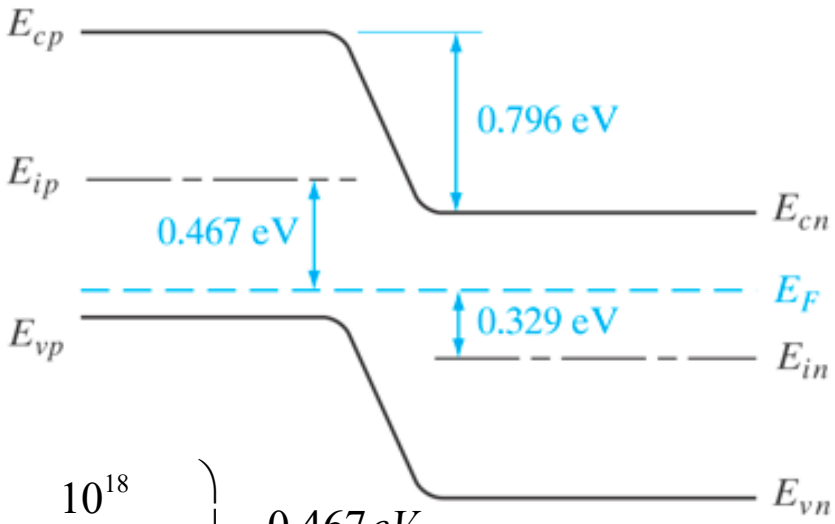
$$x_{n0} = \left\{ \frac{2\varepsilon V_0}{q} \left[\frac{N_a}{N_d (N_a + N_d)} \right] \right\}^{1/2}$$

Abrupt Junctions

Example: Abrupt Junction

Problem:

An abrupt junction in Si has $N_a=10^{18}$ cm^{-3} acceptors on one side and $N_d=5\times10^{15}$ donors on the other.



(a) Calculate the Fermi level positions at 300 K:

$$p_p = n_i e^{(E_i-E_F)/kT} \text{ so } E_{ip}-E_F = kT\ln\left(\frac{p_p}{n_i}\right) = 0.0259\ln\left(\frac{10^{18}}{(1.5\times10^{10})}\right) = 0.467eV$$

$$n_n = n_i e^{(E_F-E_i)/kT} \text{ so } E_F-E_{in} = kT\ln\left(\frac{n_n}{n_i}\right) = 0.0259\ln\left(\frac{5\times10^{15}}{(1.5\times10^{10})}\right) = 0.329eV$$

(b) Determine the contact potential:

$$qV_o = 0.467+0.329 = 0.796eV, \text{ so } V_o = 0.796V$$

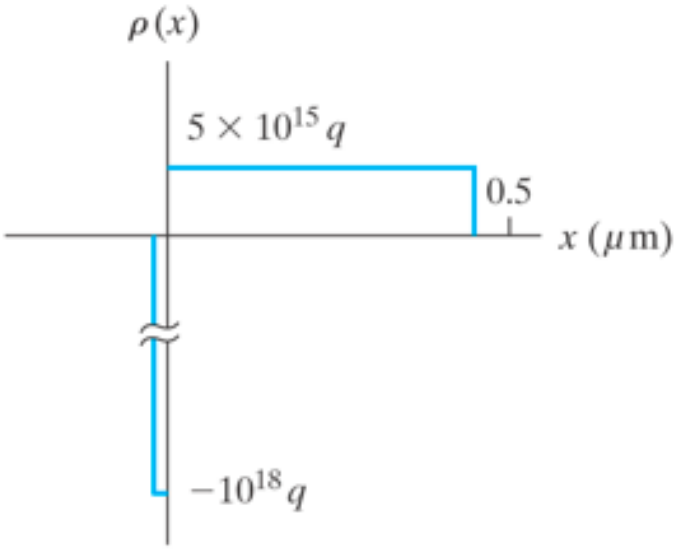
(c) Calculate the contact potential directly:

$$qV_o = kT\ln\frac{p_p}{p_n} = kT\ln\frac{N_aN_d}{n_i^2} = 0.0259\ln\left(\frac{(1\times10^{18})(5\times10^{15})}{(1.5\times10^{10})^2}\right) = 0.796eV, \text{ so } V_o = 0.796V$$

Example: Abrupt Junction [2]

Problem:

An abrupt junction in Si has $N_a=10^{18}$ cm^{-3} on one side and $N_d=5\times10^{15}$ on the other. If the junction has a diameter of 10μm:



Calculate the junction area:

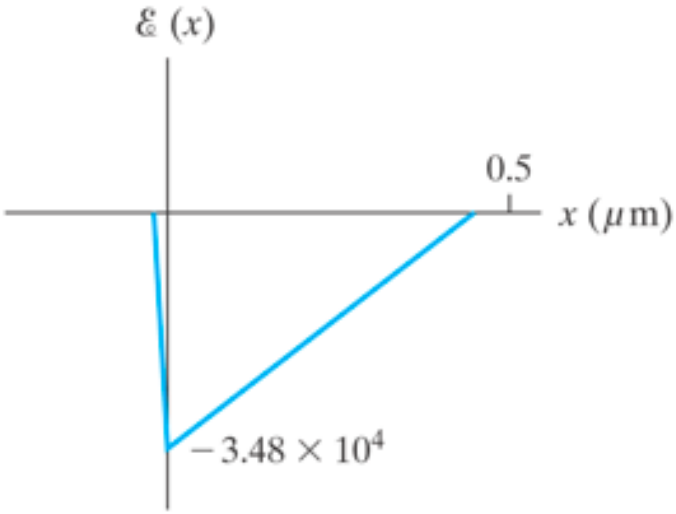
$$A=\pi(5\times10^{-4})^2=7.85\times10^{-7}\,cm^2$$

Assuming a junction in equilibrium at 300K, calculate the total depletion width:

$$W=\left[\frac{2\varepsilon V_o}{q}\left(\frac{1}{N_a}+\frac{1}{N_d}\right)\right]^{1/2}$$

$$=\left[\frac{2(11.8)(8.85\times10^{-14})(0.796)}{1.6\times10^{-19}}\left(10^{-18}+2\times10^{-16}\right)\right]^{1/2}$$

$$=0.457\,\mu m$$



Example: Abrupt Junction [3]

Problem:

An abrupt junction in Si has $N_a=10^{18}$ cm^{-3} on one side and $N_d=5\times10^{15}$ on the other. If the junction has a diameter of 10μm:

Calculate the depletion width on the n-side (x_{no}) and p-side (x_{po}):

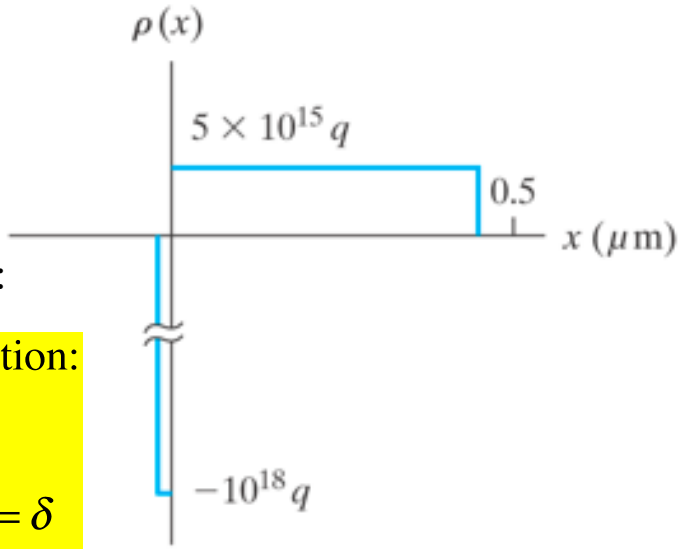
$$x_{no}=\frac{W}{1+N_d/N_a}=\frac{0.457}{1+5\times10^{-3}}=0.455\,\mu m$$

$$x_{po}=\frac{W}{1+N_a/N_d}=\frac{0.457}{1+200}=2.27\times10^{-3}\,\mu m$$

Abrupt Junction:

$$x_{no}\sim W$$

$$x_{po}\sim W\frac{N_a}{N_d}=\delta$$



Calculate the charge Q_+ and plot the charge density:

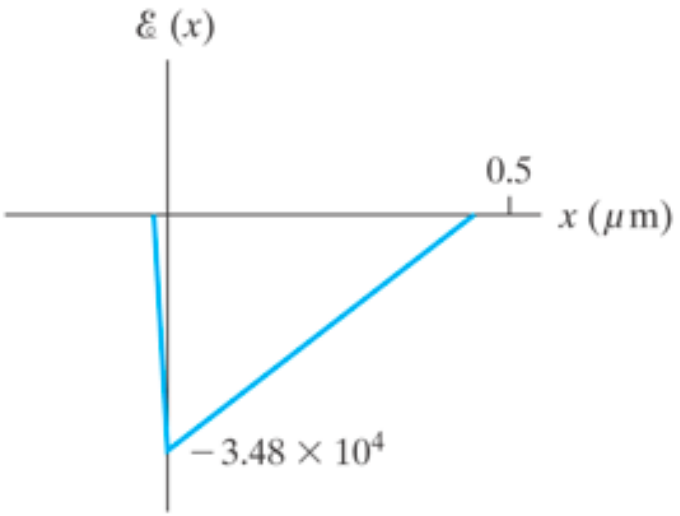
$$Q_+=qAx_{no}N_d=qAx_{po}N_a$$

$$=(1.6\times10^{-19})(7.85\times10^{-7})(2.27\times10^{11})=2.85\times10^{-14}\,C$$

Calculate and plot the electric field:

$$E_o=-\frac{q}{\varepsilon}x_{no}N_d=-\frac{q}{\varepsilon}x_{po}N_a$$

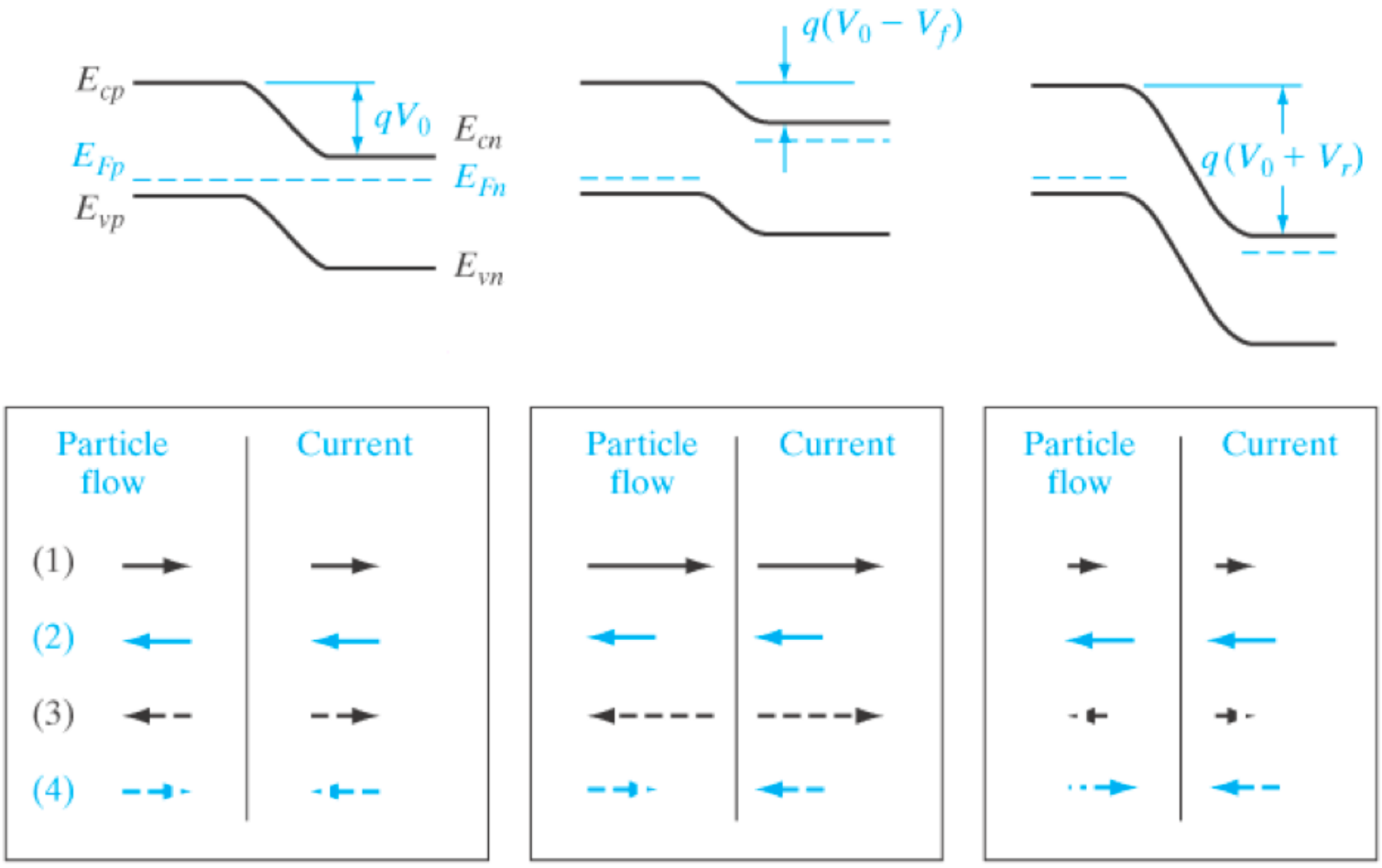
$$=\frac{1.6\times10^{-19}}{(11.8)(8.85\times10^{-14})}\left(2.27\times10^{11}\right)=-3.48\times10^{4}\,V/cm$$



pn Junctions Under Bias

A Few Comments

Electron & Hole Motion



(1) Hole diffusion
(2) Hole drift

(3) Electron diffusion
(4) Electron drift